

DSA STUDY BLUEPRINT SERIES

121. Disjoint Set Union With Rank Size Path Compression

Union-Find with Rank/Size & Path Compression

Section 10 • C++ Core Data Structures & Algorithms

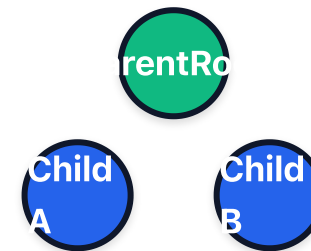
Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **DSU:** Group disjoint sets. Operations: `find` (with path compression) and `union` (by rank/size).

Memory & Execution Blueprint



C++ Implementation Reference

Standard code layout and syntax

```
int findParent(int i, vector& parent) {  
    if (parent[i] == i) return i;  
    return parent[i] = findParent(parent[i], parent);  
}  
  
void unionSets(int i, int j, vector& parent, vector& rank) {  
    int r1 = findParent(i, parent), r2 = findParent(j, parent);  
    if (r1 != r2) {  
        if (rank[r1] < rank[r2]) parent[r1] = r2;  
        else { parent[r2] = r1; if (rank[r1] == rank[r2]) rank[r1]++; }  
    }  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Find / Union	$O(\alpha(N))$ (Inverse Ackermann)

Key Practices & Gotchas

- **Important:** Path compression flattens the tree height to near-constant levels.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace Dijkstra distance relaxation updates on active vertices:

Vertex popped	Adjacent edge checked	Current distance value	New path calculation	Relaxation action
U (Dist = 2)	U → V (Weight = 3)	V (Dist = 7)	New Dist = 2 + 3 = 5	Relax: Update V (Dist = 5)

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Shortest Path choices:** BFS finds unweighted short paths; Dijkstra handles positive weights; Bellman-Ford resolves negative edges.
- **Spanning Trees:** Prim's builds MST from a start node; Kruskal's sorts all edges using disjoint sets.
- **Related LeetCode Problems:** #200 Number of Islands, #743 Network Delay Time, #787 Cheapest Flights Within K Stops.